

Neonatal Conjunctivitis

Azeem Khan, MD,* Anjali Anders, MD,[†] Maggie Cardonell, MD[‡]

*Department of Ophthalmology, University of Missouri, Columbia, MO

[†]Department of Pediatrics, University of Missouri, Columbia, MO

[‡]Harry S. Truman Memorial Veterans' Hospital, Columbia, MO

PRACTICE GAP

Neonatal conjunctivitis has infectious and noninfectious etiologies. Certain infections leading to conjunctivitis can be associated with significant morbidity, therefore, clinicians should understand the diagnosis and management of neonatal conjunctivitis.

OBJECTIVES *After completing this article, readers should be able to:*

1. Recognize the clinical features of conjunctivitis and determine its cause in neonates.
2. Describe the current treatment recommendations for conjunctivitis in neonates.

ABSTRACT

Neonatal conjunctivitis is an important cause of ocular morbidity with infectious and noninfectious origins. Common practice in the United States is to administer prophylactic antibiotic ointment at birth; however, prophylaxis is ineffective for some causes. Moreover, recognition of the etiologies that threaten systemic complications to the neonate is critical in providing timely and appropriate treatment. This review summarizes the clinical features, etiology, and treatment recommendations of various forms of neonatal conjunctivitis.

AUTHOR DISCLOSURE Drs Khan, Anders, and Cardonell have disclosed no financial relationships relevant to this article. This commentary does not contain a discussion of an unapproved/investigative use of a commercial product/device.

INTRODUCTION

Neonatal conjunctivitis, also known as *ophthalmia neonatorum*, is an ocular surface disease characterized by inflammation of the conjunctiva by infectious or noninfectious causes before 30 days of age. (1)(2)(3)(4) Physical examination may demonstrate conjunctival chemosis (edema), injection (redness), tearing, and purulent or nonpurulent discharge. (5) The eyelid margins and cornea may also be involved. (6) Potential complications of neonatal conjunctivitis include endophthalmitis, keratitis, blindness (rare in developed countries), and depending on the etiology, can have associated systemic manifestations such as pneumonia, meningitis, and sepsis. (1)

The etiology of neonatal conjunctivitis generally is chemical, bacterial, and viral in nature. Among the significant bacterial causes are maternal sexually transmitted organisms such as *Chlamydia trachomatis* and *Neisseria gonorrhoeae*, those from the maternal gastrointestinal tract, and commensal organisms found on the skin. (7) Specific

ABBREVIATIONS

CDC	Centers for Disease Control and Prevention
CLIA	Clinical Laboratory Improvement Amendments
COVID-19	coronavirus disease 2019
DFA	direct fluorescent antibody
FDA	Food and Drug Administration
HSV	herpes simplex virus
NAAT	nucleic acid amplification test
PCR	polymerase chain reaction
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
USPSTF	US Preventive Services Task Force

bacterial causes that have been identified include *Staphylococcus* (1) and *Streptococcus* (3)(8) species, *Haemophilus influenzae*, (9) *Serratia marcescens*, (10)(11) *Escherichia coli*, (12) *Moraxella catarrhalis*, (1)(3) and *Neisseria meningitidis*, among others. (13)

Neonates, and especially those born premature, are at particular risk of conjunctivitis because they spend significant time with their eyes closed, allowing for bacterial proliferation, and their immature lacrimal systems limit tear distribution and drainage. (14) Neonates also lack lymphoid tissue in the conjunctiva, have decreased immunoglobulin A, and have decreased lysozyme activity. (15) These factors limit local immune function and contribute to the severity of the ophthalmic reaction. (15) Additional risk factors of neonatal conjunctivitis include prenatal factors such as maternal vaginal infection (allowing transmission at the birth canal), maternal human immunodeficiency virus (HIV) infection, poor prenatal care, inadequate prophylaxis, premature rupture of membranes, and prolonged labor. Postnatal risk factors include premature birth, low birthweight, ocular trauma at delivery, poor hygiene during delivery, mechanical ventilation, and chemical exposure. (5)(7)(14)(16)

The incidence of neonatal conjunctivitis ranges from 1.6% to 12% of all newborns (17); the incidence and specific etiologies vary geographically and across different clinical settings. In a Finnish study of 9,600 births from 2010 to 2015, 173 neonates (1.8%) were found to have clinical conjunctivitis. (3) Of 163 of these neonates tested for *C trachomatis*, zero tested positive. In contrast, neonatal conjunctivitis causes significant ocular morbidity and blindness in developing nations. (5)(7)(18) Studies from sub-Saharan Africa suggest that *C trachomatis* may account for up to 33% of cases of neonatal conjunctivitis. (19) This variation has been attributed to differences in the prevalence of maternal infection and postnatal prophylaxis of the neonate. (7)

Although rare compared to its prevalence in other regions, *Chlamydia* has been reported as more common than *N gonorrhoeae* in the developed world; in developing nations, both chlamydial and gonococcal ocular infections are prevalent. (7) Gonococcal ophthalmia is uncommon in the United States primarily due to prenatal screening and treatment for *N gonorrhoeae* in pregnant women. (20) In 1993, the Centers for Disease Control and Prevention (CDC) began to recommend routine prenatal screening and treatment of pregnant women for *C trachomatis*. (20) Prior to this, *C trachomatis* was the most common cause of neonatal conjunctivitis in the United States. (20)

The differential diagnosis of neonatal conjunctivitis includes congenital nasolacrimal duct obstruction, ocular foreign body, preseptal or orbital cellulitis, corneal abrasion or other traumatic injury to the eye after delivery,

dacryocystitis, keratitis, subconjunctival hemorrhage, and congenital glaucoma. (21) These and other causes must be considered in the evaluation of a neonate who presents with ocular findings concerning for conjunctivitis (Table).

PROPHYLAXIS

Prophylaxis for neonatal conjunctivitis is credited to Dr Carl Credé, who introduced the practice of using silver nitrate to prevent ocular infection in the early 1880s. (7)(20) Other agents that have been used for prophylaxis include erythromycin ointment 0.5%, povidone iodine 2.5%, and tetracycline ointment 1%. (22) Gentamicin ophthalmic ointment is not recommended as it has been associated with severe ocular reactions in neonates. (23) Notably, some sources report topical silver nitrate, erythromycin, and tetracycline are only considered effective for gonococcal conjunctivitis and are ineffective as prophylaxis for chlamydial conjunctivitis. (7)(24)

The US Preventive Services Task Force (USPSTF) recommends topical ocular medication for all neonates to prevent gonococcal conjunctivitis. (25) This prophylaxis should be provided in the first hour after birth. (17) The USPSTF also recommends screening for gonorrhea in pregnant women, all sexually active women 24 years of age and younger, and older women at increased risk of infection. (26) The CDC specifically recommends neonatal ocular prophylaxis with topical erythromycin. (18)(20) Currently, in the United States, erythromycin ophthalmic ointment is the only topical prophylactic option for neonatal conjunctivitis available; tetracycline ophthalmic ointment is no longer manufactured, and silver nitrate for ophthalmic use has not been available for decades. (20)(26) Without ocular prophylaxis, the mother-to-child transmission rate of gonococcal infection ranges from 30% to 50%. (26)(27)(28) Due to concerns for rising antibiotic resistance, new candidates for prophylaxis are being developed, including fatty-acid-based formulations that in early research have demonstrated effectiveness against *N gonorrhoeae* and have been proven safe for use in animal models. (29) Monocaprin, the monoglyceride of capric acid, has shown promise as a potential treatment and prophylactic agent in ocular models and in vitro studies against *N gonorrhoeae*, *N meningitidis* (whose consequences can be lethal), and *C trachomatis*. (13)

Some countries, such as the United Kingdom, Denmark, Norway, and Sweden do not use universal ocular prophylaxis, and the Canadian Paediatric Society released a policy statement in 2015 against routine ocular topical prophylaxis with erythromycin. (24) There may be limited benefit to universal topical prophylaxis in some parts of the world due to low prevalence and effective prenatal screening. In a study of 27,556 live births at the National

Table. Common Causes of Neonatal Conjunctivitis and Differential Diagnosis. In each case, ophthalmology consultation is warranted:

Cause	Age at Onset	Features	Diagnosis	Treatment
Chemical irritant	1–2 days after exposure	Gram stain reveals leukocytes but no organisms	Recent exposure to silver nitrate, povidone-iodine scrub formulation, or other irritants	Self-limited after discontinuing exposure
<i>Neisseria gonorrhoeae</i>	2–5 days after birth	Bilateral hyperacute purulent conjunctivitis	Nucleic acid amplification testing of conjunctival swab specimen, culture	Systemic ceftriaxone or cefotaxime
<i>Chlamydia trachomatis</i>	5–14 days after birth	Watery discharge with increasing purulence	Nucleic acid amplification testing of conjunctival swab specimen, culture	Systemic erythromycin or azithromycin
Herpes simplex	5–14 days after birth; systemic disease can emerge as late as 6 weeks	Unilateral chemosis, serosanguineous discharge, vesicles around eyelids	HSV culture or PCR testing of conjunctiva, corneal epithelium, or skin vesicle scraping	Systemic acyclovir +/- topical trifluridine or ganciclovir
Corneal abrasion	After trauma to corneal epithelium	Redness, pain, tearing	Fluorescein staining	Topical antibiotic ointment, typically erythromycin
Subconjunctival hemorrhage	After mild trauma to conjunctiva	Redness without pain or discharge	Physical examination and ruling out other causes	Self-limited
Congenital glaucoma	Varies; neonatal onset possible	Light sensitivity, watery discharge, eyelid spasm, enlargement of affected eye(s), corneal striae	Intraocular pressure measurement	Medical and surgical intraocular pressure lowering interventions

Maternity Hospital in Dublin, Ireland, from 2011 to 2013, no case of neonatal gonorrhea was identified, and no topical prophylaxis was used. (30) The incidence of neonatal conjunctivitis decreased in North America in the 1980s with the advent of routine prenatal screening and treatment of sexually transmitted infections, (24) which suggests that addressing the causes of neonatal conjunctivitis in pregnant women may itself be an effective form of prophylaxis. The protection of newborns from *C trachomatis*, for example, primarily rests upon prenatal screening and treatment. This practice has decreased both chlamydial conjunctivitis and pneumonia in neonates. (20)

A report by the American Academy of Pediatrics examined the impact of parental refusal of topical prophylaxis or the possibility of a national shortage of topical ocular prophylaxis. (31) The authors found that universal screening and treatment during pregnancy is expected to lower the burden of neonatal disease, but topical prophylaxis is recommended by the USPSTF and required by law in most of the United States. Though prenatal care is the norm, 6.2% of births in the United States occur with no prenatal care. In cases without topical prophylaxis, the authors recommended assessing the parents for potential risk factors, documenting negative test results for gonorrhea in pregnancy in the infant's medical record, and

counseling the family to seek immediate medical attention for conjunctival inflammation and discharge. Testing may also be performed in the immediate peripartum setting and, if positive or pending with concern for poor follow-up, treatment may be initiated.

LABORATORY DIAGNOSIS

Diagnostic testing for neonatal conjunctivitis includes cell culture (with Gram stain and susceptibility testing) along with nonculture-based tests. (23) The American Academy of Ophthalmology recommends obtaining conjunctival culture specimens in all cases of suspected bacterial neonatal conjunctivitis. (1) The nonculture-based tests include nucleic acid amplification tests (NAATs) and direct fluorescent antibody (DFA) testing. (21) DFA is the only culture-independent US Food and Drug Administration (FDA)-approved method for detection of chlamydia from conjunctival swab specimen, but laboratories can offer NAAT for chlamydia after they have verified use of the specimen according to regulations of the Clinical Laboratory Improvement Amendments (CLIA). (23) The diagnostic standard for suspected chlamydial conjunctivitis is to obtain a specimen from an everted eyelid using a polyethylene terephthalate swab or the swab type specified by the manufacturer's kit. (2) The specimen must contain conjunctival cells, not discharge alone. (2)(23)

Similarly, although NAATs are not FDA-approved for *N gonorrhoeae* testing on a conjunctival swab specimen, they are more sensitive compared with culture. Many laboratories have met the CLIA requirement and have validated NAAT on extragenital specimens. (23) Culture is less sensitive because *N gonorrhoeae* is labile and requires rapid transport from the bedside to the laboratory to successfully grow the organism. (30) That said, culture can provide definitive diagnosis of *N gonorrhoeae*, and even Gram stain of conjunctival exudates that demonstrate an elevated white blood cell (polymorphonuclear leukocyte) count or gram-negative intracellular diplococci is sufficient for presumptive treatment. (9) Polymerase chain reaction (PCR) testing is also useful in the diagnosis of viral infections. (6) Neonates with systemic infection may be at risk for bacteremia, arthritis, sepsis, or meningitis; additional investigative studies should be performed, including but not limited to blood culture and cerebrospinal fluid testing. (18)

CHEMICAL CONJUNCTIVITIS

Traditionally, chemical conjunctivitis in the neonate was associated with silver nitrate prophylaxis. (7) Silver nitrate prophylaxis previously led to chemical conjunctivitis in approximately 50% of patients in whom it was administered. (18)(22) Conjunctivitis secondary to this noninfectious cause typically presents bilaterally within 1 to 2 days after exposure to a topical agent, (22) distinguishing this diagnosis from infectious conjunctivitis, which generally presents later. Gram stain may reveal leukocytes but no organisms. (22) Management is conservative and symptoms resolve in approximately 2 days. (22) The replacement of silver nitrate administration with other means of prophylaxis has significantly reduced the incidence of neonatal chemical conjunctivitis. As noted earlier, povidone-iodine has been used for prophylaxis, but care should be taken to use a “prep solution and not a scrub formulation, which causes significant chemical conjunctivitis and potentially corneal decompensation.” (22) Due to severe ocular reactions in neonates associated with gentamicin ophthalmic ointment, it is no longer used for prophylaxis. (23)

CHLAMYDIAL CONJUNCTIVITIS

Chlamydia is the most frequent infectious cause of neonatal conjunctivitis worldwide and is most commonly transmitted by passage through an infected birth canal. (6) *C trachomatis* is an obligate intracellular gram-negative bacterium with at least 15 serologic variants (serovars) known for causing various diseases. (23) Serovars A through C cause trachoma whereas serovars B and D through K typically cause perinatal and genital infections. (23)

Trachoma causes up to 3% of the world's blindness, making it the leading cause of infectious blindness worldwide. (23) Trachoma presents with mucopurulent discharge that progresses to scarring and mechanical damage of the eyelids, conjunctiva, and cornea. (2)(6) It is transmitted by the transfer of ocular discharge and is most commonly found in resource-limited parts of the world. (23) It is typically contracted outside of the neonatal period and is rarely observed in the United States. (2)(23)

Symptoms of neonatal chlamydia conjunctivitis, in contrast, typically present 5 to 14 days after birth (6)(7) beginning with watery discharge which becomes more copious and purulent over time. Many cases are mild and self-limited but may progress to include chemosis, eyelid swelling, conjunctival papillae, pseudomembrane formation, peripheral pannus, and corneal involvement. (7)

Chlamydial infection in pregnancy is an important cause of morbidity with potential complications that include premature rupture of membranes, preterm labor and delivery, low birthweight, and fetal loss. (19) Premature birth places the neonate at greater risk for morbidity and mortality. Postnatally, *C trachomatis* can lead to pneumonia (typically presents outside the neonatal period at 4–11 weeks of age) (32) in addition to conjunctivitis. (19) It is estimated that in pregnant women with active untreated *C trachomatis*, 30% to 50% of their neonates will develop conjunctivitis. (19) Up to 20% of neonates exposed to *C trachomatis* may develop pneumonia; of these, approximately 50% have a history of conjunctivitis. (6)

The primary means of preventing neonatal chlamydial infection is screening and treatment of pregnant women, a practice that has been implemented in the United States and other countries. (19) Neonatal chlamydial conjunctivitis may be treated with systemic erythromycin 50 mg/kg per day orally in 4 divided doses for 14 days. (23) The efficacy of erythromycin is approximately 80%, so a second course of treatment may be necessary. (23) Systemic use of erythromycin in neonates is associated with an increased risk for pyloric stenosis. (33) Azithromycin is an alternative treatment option that is typically dosed at 20 mg/kg as a single daily dose for 3 days, (23) which may improve compliance due to less frequent dosing, but it is thought to be less effective than erythromycin. (33) Neonates diagnosed with chlamydial conjunctivitis should be assessed for gonococcal infection with appropriate management as indicated.

GONOCOCCAL CONJUNCTIVITIS

N gonorrhoeae, a gram-negative diplococcus, is responsible for approximately 0.4 cases of neonatal conjunctivitis per 100,000 live births in the United States. Although a rare

cause of conjunctivitis, it is associated with significant morbidity. (25) Transmission typically occurs during delivery with exposure to infectious vaginal secretions; the maternal cervix and urethra may serve as bacterial reservoirs. (9) Even with caesarian section, vertical transmission of *N gonorrhoeae* is possible. The CDC reported a rate of 179 cases of gonorrhea per 100,000 population in 2019; this represents an increase of 82.6% over the historic low in 2009. (23) Pregnant women with *N gonorrhoeae* infection are at a higher risk of having adverse birth outcomes including premature rupture of membranes, preterm birth, low birthweight, and perinatal mortality. (34)

Gonococcal conjunctivitis typically presents as bilateral, hyperacute conjunctivitis 2 to 5 days after birth. (6)(7) Even with prophylaxis (typically with erythromycin ophthalmic ointment in the United States), 10% of neonates exposed to gonorrhea during delivery may develop gonococcal conjunctivitis. (35) *N gonorrhoeae* has the potential to penetrate intact corneal epithelium, leading to corneal ulceration, which can progress to perforation and endophthalmitis. (7) Systemic manifestations such as rhinitis, arthritis, sepsis, and meningitis are also possible, highlighting the importance of screening and treatment of gonococcal infections. (7) If confirmatory testing is not available and the clinician has high suspicion for gonorrhea and/or chlamydia, treatment against both should be started to avoid complications. (18)

First-line treatment of gonococcal conjunctivitis consists of ceftriaxone. Non-disseminated infections may be treated with a single intramuscular or intravenous injection of ceftriaxone 25 to 50 mg/kg, not to exceed 125 mg and contraindicated if receiving simultaneous intravenous calcium. (6)(9)(23) An alternative treatment is a single 100 mg/kg intramuscular or intravenous dose of cefotaxime. (6)(23) In either case, saline irrigation of the ocular surface should be performed. (23) If disseminated infection is suspected, an infectious disease specialist should be consulted. Ophthalmology consultation is indicated in either case, and assessment for concomitant infections such as HIV, syphilis, and chlamydia should be performed. (21)

VIRAL CONJUNCTIVITIS

A number of viruses have been implicated in neonatal conjunctivitis including herpes simplex virus (HSV), adenovirus, rhinovirus, and coronavirus, among others. (3) HSV conjunctivitis is typically associated with vesicular lesions on the eyelid skin. (6)(7) HSV may infect a neonate during vaginal delivery or even by ascending intrauterine infection if the pregnant woman has genital HSV. (22) The risk of infection is significantly higher with maternal primary as opposed to reactivated infection (25%–60% versus 2%, respectively). (22) Neonatal

infection can also occur by direct contact with caregivers affected by herpetic whitlow or herpes labialis. (22) Clinically apparent disease usually emerges 5 to 14 days after exposure; systemic disease is also a possibility and can manifest as late as 6 weeks after exposure. (22) Although herpes simplex causes less than 1% of neonatal conjunctivitis, it should be included in the differential for neonates presenting with unilateral chemosis, serosanguineous discharge, or vesicular lesions surrounding the eyelids. (18) Complications are potentially lethal and may include disseminated and central nervous system disease. (18) Laboratory testing is warranted and should include HSV culture or PCR testing of the conjunctiva, corneal epithelium, or skin vesicle scraping, cerebrospinal fluid and serum testing with HSV PCR, and liver function tests. (22) Infants suspected to have herpes simplex ocular infection should be examined by an ophthalmologist as infection may lead to corneal involvement, cataracts, and/or chorioretinitis. (21) Neonatal HSV infection is treated with intravenous acyclovir 60 mg/kg per day in 3 divided doses for 14 days, or 21 days in the setting of meningoencephalitis or disseminated disease. (22)(23) Adjunctive therapy with topical trifluridine 1% or ganciclovir 0.15% may also be considered. (22)(23) Enteral acyclovir suppression is continued for 6 months after the completion of intravenous therapy in infants surviving neonatal HSV disease of any classification. (23)

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), the cause of coronavirus disease 2019 (COVID-19), has been reported as a cause of conjunctivitis. (15)(36) One case report described a neonate who presented with acute mucopurulent discharge, subconjunctival hemorrhage, and palpebral injection. (15) Laboratory diagnosis was negative for bacterial growth, *C trachomatis*, *N gonorrhoeae*, and HSV. Nasopharyngeal and conjunctival PCR testing for SARS-CoV-2 were positive. These patients may also have other associated symptoms of COVID-19.

Adenovirus is a common cause of viral conjunctivitis that may also affect neonates. It is associated with petechial hemorrhage and occasionally large subconjunctival hemorrhages and lymphadenopathy. (17) These neonates may have associated respiratory symptoms.

OTHER CAUSES

Other bacterial causes of neonatal conjunctivitis include *Staphylococcus aureus*, *Streptococcus pneumoniae*, *H influenzae*, *E coli* (Fig), *Klebsiella* species, *Pseudomonas aeruginosa*, *N meningitidis*, and others. (6)(16)(18)(37) *S marcescens*, an aerobic gram-negative bacillus, has been associated with nosocomial infections in NICUs and may present with conjunctivitis, urinary tract infection, pneumonia, sepsis,

and/or meningitis. (38)(39) Hospital-acquired conjunctivitis is generally defined to occur 48 hours or more after hospitalization; it is not always certain whether the infection was contracted during or after birth. (10) Predictive factors include low birthweight, use of a ventilator or nasal continuous positive airway pressure, and ophthalmologic examination. (14) A study of 1,492 neonates admitted to a level III neonatal unit in Portugal showed a 4% incidence of hospital-acquired conjunctivitis, in which the most common pathogens were *S marcescens*, *E coli*, and *P aeruginosa*. (10) Other studies have found coagulase-negative staphylococci, *S aureus*, and *Klebsiella* among the predominant pathogens. (14)

Empiric antibiotic therapy for bacterial conjunctivitis includes topical erythromycin as well as fluoroquinolones such as ciprofloxacin, gatifloxacin, and moxifloxacin. (16) Sanfilippo et al (1) compared besifloxacin 0.6% and gatifloxacin 0.3% in a multicenter, randomized, double-masked, parallel group study of 33 neonates and found that clinical resolution and bacterial eradication were comparable between the groups, with neither group showing any treatment-related adverse effects. Moxifloxacin and ciprofloxacin have previously been shown to be effective. (1) A pooled analysis of topical fluoroquinolone use to treat bacterial conjunctivitis revealed similar cure rates among neonates and older children, suggesting that these medications are reasonable for certain cases of neonatal bacterial conjunctivitis. (16) Notably, fluoroquinolone resistance has been increasing among certain bacterial species. Pediatric infectious disease specialist consultation may be indicated, for instance, in cases of methicillin-resistant *S aureus* or pseudomonal conjunctivitis.

Linear immunoglobulin A bullous dermatosis is an autoimmune blistering disease leading to mucosal and skin involvement including conjunctivitis as well as oral and nasal erosions. Histopathology demonstrates linear deposition of immunoglobulin A at the cutaneous basement membrane zone. (40) The mean age at onset is between 4 and 5 years, though presentations in neonates

have been reported, including in a 3 day old with vesicles and tense blisters on the scrotum, face, and neck area along with bilateral mucopurulent conjunctivitis. (40)

A case report of neonatal Kawasaki disease described a 15-day-old male infant who presented with poor feeding, irritability, fever, and rash. (41) By day 6 of illness, the infant demonstrated a bilateral nonpurulent conjunctivitis. Antibiotic and antiviral therapy were ineffective. The patient met 3 of 5 criteria for Kawasaki disease and, with no other diagnosis apparent, was treated for incomplete Kawasaki disease with intravenous immunoglobulin and demonstrated substantial improvement in symptoms within 48 hours. This disease should be considered in neonates who display systemic symptoms associated with conjunctivitis.

Summary

- Neonatal conjunctivitis is a disease entity with both infectious and noninfectious etiologies.
- Potential ocular complications of neonatal conjunctivitis include keratitis, endophthalmitis, and blindness. Systemic manifestations of certain causes may include pneumonia, meningitis, and sepsis.
- Recommendations for prophylaxis vary around the world, and the approach may be influenced by disease incidence and the availability of prenatal screening. Topical erythromycin is often used to prevent gonococcal conjunctivitis.
- Less common causes of neonatal conjunctivitis are important to consider when the clinical picture suggests them.
- It is critical to determine the cause of neonatal conjunctivitis to provide appropriate treatment and, in some cases, prevent significant morbidity and mortality.



Figure. External photograph of a neonate with *Escherichia coli* conjunctivitis. Notice the eyelid erythema, mucopurulent discharge, and conjunctival inflammation.

American Board of Pediatrics Neonatal-Perinatal Content Specifications

1. Know the causes and clinical and laboratory features of acute neonatal infections of the eyes, including ophthalmia neonatorum.
2. Know the management and complications of acute neonatal infections of the eyes, including ophthalmia neonatorum.

References

- Sanfilippo CM, Allaire CM, DeCory HH. Besifloxacin ophthalmic suspension 0.6% compared with gatifloxacin ophthalmic solution 0.3% for the treatment of bacterial conjunctivitis in neonates. *Drugs R D*. 2017;17(1):167–175
- Mishori R. Chlamydia trachomatis infections: screening, diagnosis, and management. *Am Fam Physician*. 2012;86(12):1127–1132
- Honkila M, Renko M, Ikäheimo I, Pokka T, Uhari M, Tapiainen T. Aetiology of neonatal conjunctivitis evaluated in a population-based setting. *Acta Paediatr*. 2018;107(5):774–779
- Tang S, Li M, Chen H, Ping G, Zhang C, Wang S. A chronological study of the bacterial pathogen changes in acute neonatal bacterial conjunctivitis in southern China. *BMC Ophthalmol*. 2017;17(1):174
- Suhas P, Vishnu S, Muthayya M. Pathogenic bacteria and their antibiotic sensitivity in ophthalmia neonatorum. *Oman J Ophthalmol*. 2021;14(2):85–87
- Azari AA, Arabi A. Conjunctivitis: a systematic review. *J Ophthalmic Vis Res*. 2020;15(3):372–395
- Mallika P, Asok T, Faisal H, Aziz S, Tan A, Intan G. Neonatal conjunctivitis: a review. *Malays Fam Physician*. 2008;3(2):77–81
- Soto-Noguerón A, Carnalla-Barajas MN, Solórzano-Santos F, et al. Streptococcus pneumoniae as cause of infection in infants less than 60 days of age: serotypes and antimicrobial susceptibility. *Int J Infect Dis*. 2016;42:69–73
- Mayor MT. Diagnosis and management of gonococcal infections. *Am Fam Physician*. 2012;86(10):931–938
- Dias C, Gonçalves M, João A. Epidemiological study of hospital-acquired bacterial conjunctivitis in a level III neonatal unit. *ScientificWorldJournal*. 2013;2013:163582
- Montagnani C, Cocchi P, Lega L, et al. Serratia marcescens outbreak in a neonatal intensive care unit: crucial role of implementing hand hygiene among external consultants. *BMC Infect Dis*. 2015;15(1):11
- Maalouli WM, Pitt MB. The eyes have it: an unusual case of Escherichia coli ophthalmia neonatorum or a shifting landscape? *SAGE Open Medical Case Reports*. 2017;5:2050313X1774590
- Dolange V, Churchward CP, Christodoulides M, Snyder LAS. The growing threat of gonococcal blindness. *Antibiotics (Basel)*. 2018;7(3):E59
- Haas J, Larson E, Ross B, See B, Saiman L. Epidemiology and diagnosis of hospital-acquired conjunctivitis among neonatal intensive care unit patients. *Pediatr Infect Dis J*. 2005;24(7):586–589
- Mechel E, Trinh M, Kodsí S, Hymowitz M, Kainth MK, Lee AM. Ophthalmia neonatorum as the presenting sign of SARS-CoV-2. *J AAPOS*. 2021;25(4):230–231
- Sun H, Temeck JW, McCune SK, Chambers WA. Using historical data to support approval of drug products for ophthalmia neonatorum: innovative approach toward evidence-based medicine. *JAMA Ophthalmol*. 2021;139(2):214–217
- Matejcek A, Goldman RD. Treatment and prevention of ophthalmia neonatorum. *Can Fam Physician*. 2013;59(11):1187–1190
- Castro Ochoa KJ, Mendez MD. Ophthalmia neonatorum. In: *StatPearls*. Treasure Island, FL: StatPearls Publishing; 2021. <http://www.ncbi.nlm.nih.gov/books/NBK551572/Accessed June 13, 2022>
- Adachi KN, Nielsen-Saines K, Klausner JD. Chlamydia trachomatis screening and treatment in pregnancy to reduce adverse pregnancy and neonatal outcomes: a review. *Front Public Health*. 2021;9:531073 10.3389/fpubh.2021.531073
- Smith-Norowitz TA, Ukaegbu C, Kohlhoff S, Hammerschlag MR. Neonatal prophylaxis with antibiotic containing ointments does not reduce incidence of chlamydial conjunctivitis in newborns. *BMC Infect Dis*. 2021;21(1):270
- Makker K, Nassar GN, Kaufman EJ. Neonatal conjunctivitis. In: *StatPearls*. Treasure Island, FL: StatPearls Publishing; 2021. <http://www.ncbi.nlm.nih.gov/books/NBK441840/Accessed June 13, 2022>
- Lyons CJ, Lambert SR. *Taylor and Hoyt's Pediatric Ophthalmology and Strabismus E-Book*. Philadelphia, PA: Elsevier Health Sciences; 2016
- Committee on Infectious Diseases, American Academy of Pediatrics; Kimberlin DW, Barnett ED, Lynfield R, Sawyer MH, eds. *Red Book: 2021–2024 Report of the Committee on Infectious Diseases*. 32nd ed. Itasca, IL: American Academy of Pediatrics; 2021
- Moore DL, MacDonald NE. Société canadienne de pédiatrie, comité des maladies infectieuses et d'immunisation. La prévention de la conjonctivite néonatale. *Paediatr Child Health*. 2015;20(2):97–100 doi: 10.1093/pch/20.2.97
- Guirguis-Blake JM, Evans CV, Rushkin M. Ocular prophylaxis for gonococcal ophthalmia neonatorum: updated evidence report and systematic review for the US Preventive Services Task Force. *JAMA*. 2019;321(4):404–406
- Ocular prophylaxis for gonococcal ophthalmia neonatorum: recommendation statement. *Am Fam Physician*. 2019;100(4):236–238
- Kreisel K, Weston E, Braxton J, Llata E, Torrone E. Keeping an eye on chlamydia and gonorrhea conjunctivitis in infants in the United States, 2010–2015. *Sex Transm Dis*. 2017;44(6):356–358
- Justel M, Alexandre I, Martinez P, et al. Vertical transmission of bacterial eye infections, Angola, 2011–2012. *Emerg Infect Dis*. 2015;21(3):471–473
- Churchward CP, Alany RG, Kirk RS, Walker AJ, Snyder LAS. Prevention of ophthalmia neonatorum caused by Neisseria gonorrhoeae using a fatty acid-based formulation. *mBio*. 2017;8(4):e00534-17. doi:10.1128/mBio.00534-17
- McAnena L, Knowles SJ, Curry A, Cassidy L. Prevalence of gonococcal conjunctivitis in adults and neonates. *Eye (Lond)*. 2015;29(7):875–880
- Nolt D, O'Leary ST, Aucott SW. Risks of infectious diseases in newborns exposed to alternative perinatal practices. *Pediatrics*. 2022;149(2):e2021055554 doi: 10.1542/peds.2021-055554
- Bekler C, Kultursay N, Ozacar T, Sayiner A, Yalaz M, Akisu M. Chlamydial infections in term and preterm neonates. *Jpn J Infect Dis*. 2012;65(1):1–6
- Zikic A, Schünemann H, Wi T, Lincetto O, Broutet N, Santesso N. Treatment of neonatal chlamydial conjunctivitis: a systematic review and meta-analysis. *J Pediatric Infect Dis Soc*. 2018;7(3):e107–e115
- Vallély LM, Egli-Gany D, Wand H, et al. Adverse pregnancy and neonatal outcomes associated with Neisseria gonorrhoeae: systematic review and meta-analysis. *Sex Transm Infect*. 2021;97(2):104–111
- Costumbrado J, Ng DK, Ghassemzadeh S. Gonococcal conjunctivitis. In: *StatPearls*. Treasure Island, FL: StatPearls Publishing; 2021. <http://www.ncbi.nlm.nih.gov/books/NBK459289/Accessed June 13, 2022>
- Valente P, Iarossi G, Federici M, et al. Ocular manifestations and viral shedding in tears of pediatric patients with coronavirus disease 2019: a preliminary report. *J AAPOS*. 2020;24(4):212–215
- Kretz CB, Bergeron G, Aldrich M, et al. Neonatal conjunctivitis caused by Neisseria meningitidis US Urethritis Clade, New York, USA, August 2017. *Emerg Infect Dis*. 2019;25(5):972–975

38. Cristina ML, Sartini M, Spagnolo AM. *Serratia marcescens* infections in neonatal intensive care units (NICUs). *Int J Environ Res Public Health*. 2019;16(4):610
39. Yeo KT, Octavia S, Lim K, et al. *Serratia marcescens* in the neonatal intensive care unit: A cluster investigation using molecular methods. *J Infect Public Health*. 2020;13(7):1006–1011
40. Romani L, Diociaiuti A, D'Argenio P, et al. A case of neonatal linear IgA bullous dermatosis with severe eye involvement. *Acta Derm Venereol*. 2015;95(8):1015–1017
41. Altammar F, Lang B. Kawasaki Disease in the neonate: case report and literature review. *Pediatr Rheumatol Online J*. 2018;16(1):43



1. Neonatal conjunctivitis can be an infectious or noninfectious condition occurring in the first month of age. Preterm infants are at comparatively higher risk of developing neonatal conjunctivitis. Which of the following is a mechanism for this increased risk in preterm infants?
 - A. The common use of systemic antibiotics, especially aminoglycosides, leading to noninfectious side effect of eye drainage.
 - B. Significant time spent with eyes closed, allowing for bacterial proliferation and immature lacrimal systems limiting tear drainage.
 - C. Abundance of nonfunctional lymphoid tissue in conjunctiva.
 - D. Decreased complement activation in the eye leading to decreased immune function.
 - E. Relatively high lysozyme activity leading to chemical irritation.
2. Prophylaxis for neonatal conjunctivitis is a standard practice to prevent ocular infection. Which of the following statements regarding this prophylaxis is correct?
 - A. Gentamicin is considered the safest agent, with no side effects ever observed in neonates, but may only be about 50% effective against infectious agents.
 - B. Topical silver nitrate is considered as the optimal first-line agent in most settings.
 - C. Agents such as erythromycin and tetracycline are effective only for chlamydial conjunctivitis, not gonococcal conjunctivitis.
 - D. The US Preventive Task Force recommends topical ocular medication for all neonates to prevent gonococcal conjunctivitis.
 - E. The optimal timing of prophylaxis is after the first hour and before 48 hours after birth.
3. A term newborn is vigorous and active at birth without abnormalities noted on physical examination. The mother requests that no ophthalmic antibiotics are applied due to personal preference. What is the mother-to-child transmission rate of gonococcal infection?
 - A. 1%–5%.
 - B. 10%–15%.
 - C. 30%–50%.
 - D. 75%–80%.
 - E. >90%.
4. A term newborn infant is identified as having chlamydial conjunctivitis. Which of the following is the most appropriate treatment?
 - A. Systemic erythromycin administered orally for 14 days, with a potential second course if the first course is not efficacious.
 - B. Intravenous gentamicin for 10 days.
 - C. Ophthalmic antibiotic such as polymyxin B for 10 days.
 - D. Close observation without antibiotics if the condition resolves within 1 to 2 weeks.
 - E. Intravenous azithromycin until C-reactive protein returns to normal.

REQUIREMENTS: Learners can take *NeoReviews* quizzes and claim credit online only at: <https://publications.aap.org/neoreviews>.

To successfully complete 2022 *NeoReviews* articles for AMA *PRA Category 1 Credit™*, learners must demonstrate a minimum performance level of 60% or higher on this assessment. If you score less than 60% on the assessment, you will be given additional opportunities to answer questions until an overall 60% or greater score is achieved.

This journal-based CME activity is available through Dec. 31, 2024, however, credit will be recorded in the year in which the learner completes the quiz.



2022 *NeoReviews* is approved for a total of 30 Maintenance of Certification (MOC) Part 2 credits by the American Board of Pediatrics (ABP) through the AAP MOC Portfolio Program. *NeoReviews* subscribers can claim up to 30 ABP MOC Part 2 points upon passing 30 quizzes (and claiming full credit for each quiz) per year. Subscribers can start claiming MOC credits as early as October 2022. To learn how to claim MOC points, go to: <https://publications.aap.org/journals/pages/moc-credit>.

5. A newborn term infant is born vaginally after spontaneous labor and appears healthy. She rooms in with the mother and is breastfeeding, exhibiting a normal voiding and stooling pattern. At 2 days of age, she presents with bilateral, hyperacute conjunctivitis. She had received erythromycin ophthalmic ointment prophylaxis after delivery. Even with prophylaxis, what is the proportion of neonates exposed to gonorrhea during delivery who may develop gonococcal conjunctivitis?
- A. <1%.
 - B. 10%.
 - C. 25%.
 - D. 50%.
 - E. 90%.